

Network Scan Report (Home Lab)

1. Project Title

Basic Network Scanning and Security Analysis using Nmap

2. Objective

The objective of this project is to perform a basic network scan on a local system to identify open ports, active services, and possible security risks, and to understand how attackers may exploit weak configurations.

3. Tools Used

- Kali Linux
- Nmap (Network Scanner)
- Local Network (Home Lab Environment)

4. Methodology

1. Identified local system IP address using network commands.
2. Performed basic port scanning using Nmap.
3. Performed service version detection scan.
4. Scanned the local network to detect connected devices.
5. Analyzed scan results for potential security issues.

5. Commands Used

```
ip a
```

```
nmap (target-ip)
```

```
nmap -sV (target-ip)
```

```
nmap 192.168.1.0/24
```

6. Results

- Multiple open ports were detected on the target system.
- Common services such as SSH, HTTP, or others may be running depending on the system.
- Active devices were discovered on the local network.
- Service versions were identified for further analysis.

7. Risks Identified

1. Open Ports Risk

Some ports were found open on the system. Open ports can allow unauthorized users to attempt connections and access system services if not properly secured.

2. Outdated Services Risk

If any running services are outdated, they may contain known vulnerabilities that attackers can exploit to gain access to the system.

3. Unnecessary Services Risk

Running unnecessary services increases the attack surface and gives attackers more opportunities to find weaknesses in the system.

8. Recommendations

- Close all unused ports and disable unnecessary services
- Keep all software and services updated regularly
- Use a firewall to control incoming and outgoing traffic
- Monitor network activity for unusual behavior
- Apply strong authentication methods (strong passwords, SSH keys)

9. Conclusion

This project helped in understanding basic network scanning using Nmap and how attackers can identify vulnerabilities through open ports and services. It also highlights the importance of securing systems by minimizing exposed services and keeping software updated.

10. Learning Outcome

- Understood basic network scanning concepts
- Learned how to use Nmap for reconnaissance
- Identified potential security risks in a system
- Gained experience in writing a basic security report

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Thanks